Delivery and Shipment Management System

1. Purpose and System Scope

The Delivery and Shipment Management System manages the full life of a shipment, from booking to proof of delivery, for a delivery company and its customers.

The system is responsible for the following functions:

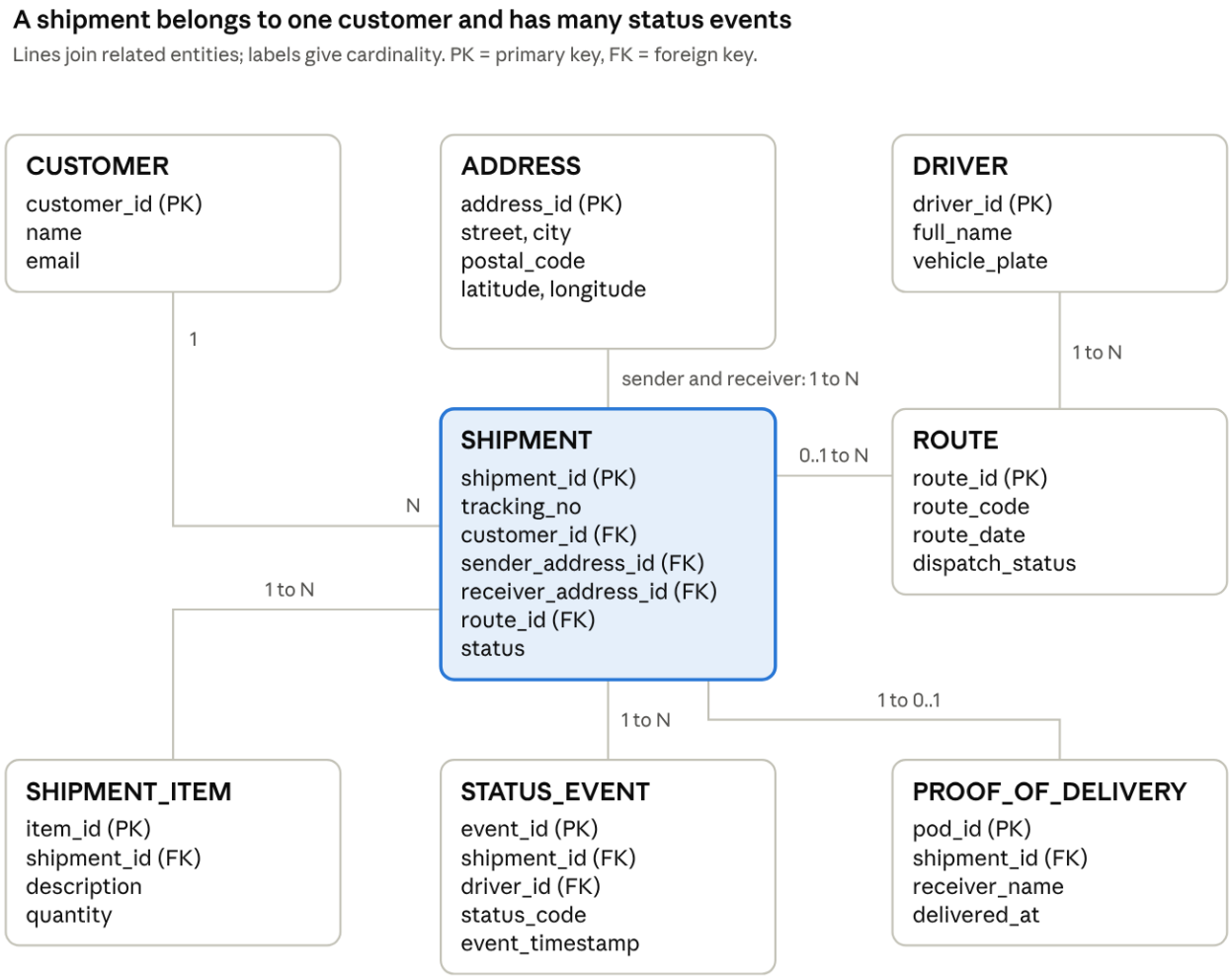
The system shall support shipment booking by allowing users to create a shipment with the sender, receiver, package details, and pickup and delivery addresses. It shall provide dispatch and assignment features that let dispatchers assign shipments to a delivery route and to a driver or courier. For tracking, the system shall record every status change and display the current status and full history to both staff and customers. It shall capture proof of delivery, including the receiver's name, a signature or photo, and the delivery timestamp. The system shall also handle exceptions by recording failed deliveries, damaged items, returns, and reschedules. To support billing, it shall compute shipment charges based on weight, distance, and service type. For reporting, it shall produce daily dispatch, on-time delivery, and exception reports. Finally, it shall provide user and access management that controls what each role (admin, dispatcher, driver, and customer) can view and change.

Out of scope for this version:

- Route optimization algorithms, which are handled by an external routing service.
- Payment gateway processing, which is handled by an external payment provider.
- Warehouse inventory management.

2. Entity Relationship Diagram

The core data model has eight entities. Each shipment belongs to one customer and moves through many status events.



Entities and attributes

Entity	Attributes	Primary key	Foreign keys
CUSTOMER	customer_id, name, contact_no, email, billing_address	customer_id	none

Entity	Attributes	Primary key	Foreign keys
ADDRESS	address_id, street, city, province, postal_code, latitude, longitude	address_id	none
SHIPMENT	shipment_id, tracking_no, customer_id, sender_address_id, receiver_address_id, service_type, status, weight_kg, declared_value, created_at	shipment_id	customer_id, sender_address_id, receiver_address_id, route_id
SHIPMENT_ITEM	item_id, shipment_id, description, quantity, dimensions	item_id	shipment_id
ROUTE	route_id, route_code, route_date, dispatch_status	route_id	none
DRIVER	driver_id, full_name, license_no, vehicle_plate, status	driver_id	none
STATUS_EVENT	event_id, shipment_id, driver_id, status_code, remarks, event_timestamp	event_id	shipment_id, driver_id
PROOF_OF_DELIVERY	pod_id, shipment_id, receiver_name, signature_file, photo_file, delivered_at	pod_id	shipment_id

Relationships and cardinality

Relationship	From	To	Cardinality	Meaning
CUSTOMER books SHIPMENT	CUSTOMER	SHIPMENT	1 to many	A customer books many shipments; each shipment has one customer.
ADDRESS is sender of SHIPMENT	ADDRESS	SHIPMENT	1 to many	An address can be the sender for many shipments.
ADDRESS is receiver of SHIPMENT	ADDRESS	SHIPMENT	1 to many	An address can be the receiver for many shipments.
SHIPMENT contains SHIPMENT_ITEM	SHIPMENT	SHIPMENT_ITEM	1 to many	A shipment has one or more items.
ROUTE carries SHIPMENT	ROUTE	SHIPMENT	0 to many	A route is optional until dispatch; a shipment is on at most one route at a time.
DRIVER delivers ROUTE	DRIVER	ROUTE	1 to many	A driver runs many routes over time; each route has one driver.
SHIPMENT has STATUS_EVENT	SHIPMENT	STATUS_EVENT	1 to many	Every status change is stored as an event.
SHIPMENT has PROOF_OF_DELIVERY	SHIPMENT	PROOF_OF_DELIVERY	1 to 0..1	A delivered shipment has one proof of delivery.

Notes

- SHIPMENT.route_id is a nullable foreign key to ROUTE, so a shipment is on at most one route at a time.

3. Business Rules

Shipment status

A shipment moves through these statuses in order. Only the transitions shown are allowed.

1. BOOKED: shipment created, not yet picked up.
2. PICKED_UP: driver scanned the parcel at the sender address.
3. IN_TRANSIT: parcel is at a hub or on a route.
4. OUT_FOR_DELIVERY: driver has the parcel on the delivery route.
5. DELIVERED: proof of delivery recorded. Final status.
6. FAILED_ATTEMPT: delivery was tried and failed; returns to OUT_FOR_DELIVERY or moves to RETURNED.
7. RETURNED: parcel is returned to the sender. Final status.
8. CANCELLED: booking cancelled before PICKED_UP. Final status.

Validation rules

- Every shipment must have a sender address, a receiver address, and at least one item.
- weight_kg must be greater than 0 and no more than 70 kg for standard service.
- declared_value must be 0 or greater; a shipment with a declared value above 50,000 PHP requires dispatcher approval.
- tracking_no is unique across all shipments and is generated by the system, never typed by staff.
- A status change must include a timestamp and the user or driver who made it.

Dispatch and delivery rules

- A shipment can be assigned to a route only when its status is BOOKED or PICKED_UP.
- A shipment can be on one route at a time. Reassigning it removes it from the previous route.
- A driver can carry at most 60 shipments per route.
- A PROOF_OF_DELIVERY record is required before a shipment can move to DELIVERED.

- A shipment in FAILED_ATTEMPT can be attempted up to 3 times. After the third failed attempt, it moves to RETURNED unless the customer requests a reschedule.

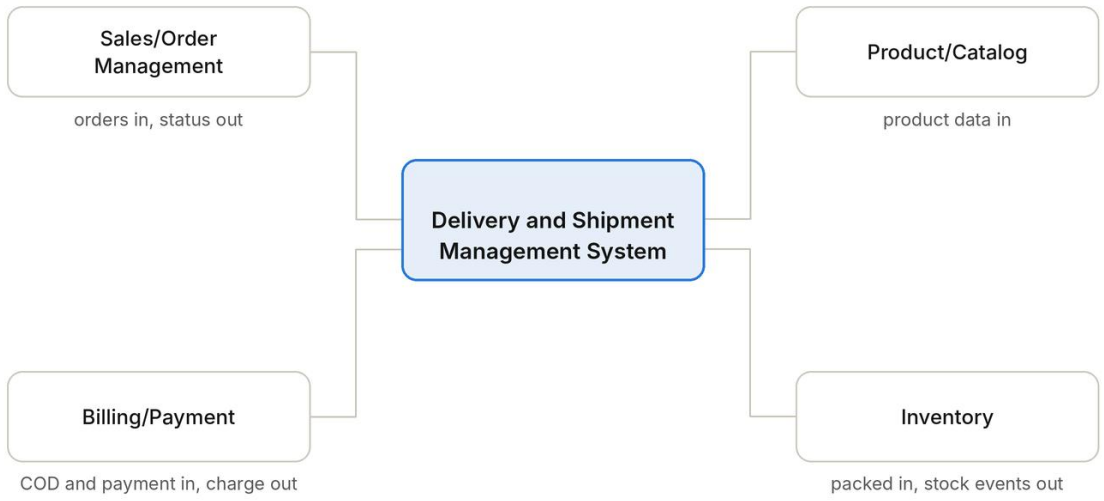
Billing rules

- Charges are computed from the service type, the chargeable weight (the higher of actual or volumetric weight), and the distance zone.
- Cancelled shipments before PICKED_UP are not billed.

4. Integration Requirements

The Delivery and Shipment Management System is one of five modules of the e-commerce ERP. The other four are Sales/Order Management, Inventory, Product/Catalog, and Billing/Payment. This section defines what the delivery module receives from and provides to each of them, so that all five modules work from the same data. Each row lists what the delivery module receives and what it provides.

The delivery module exchanges data with four ERP modules



ERP module	Receives from it	Provides to it	Method	Frequency
Sales/Order Management	Order ID, customer, receiver name and contact, delivery address, order items (product ID and quantity), service type, declared value, order status (confirmed or cancelled)	Shipment ID, tracking number, every status change, delivery confirmation with proof of delivery, failed delivery and return notices		Real time
Inventory	Packed-and-ready signal and pickup location (warehouse address) for each order item	Pickup event so stock is deducted; return event with item condition (good or damaged) so stock is restored or written off; cancellation event so reserved stock is released		Real time, on each status event
Product/Catalog	Product ID or SKU, name, weight, dimensions, and fragile or special-handling flag for each shipment item	Nothing is written back; product IDs are only referenced on shipment items		On demand, at booking

ERP module	Receives from it	Provides to it	Method	Frequency
Billing/Payment	Payment status of the order (paid, pending, or cash on delivery), COD amount to collect, invoice reference	Shipping charge (shipment ID, order ID, service type, chargeable weight, zone, amount); COD amount collected at delivery; no-charge notice for cancelled shipments; refund trigger for returns		Charge at booking; final charge at pickup; COD at delivery